AI/ML intern DAY 7

TASK — Body Position and Angle Detection System

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1. Introduction

This project focuses on developing an AI-Based Body Position and Angle Detection System using Artificial Intelligence, Computer Vision, MediaPipe, and OpenCV.

The system analyses uploaded human body images and performs:

- body landmark detection
- skeletal visualization
- body angle estimation
- body posture analysis
- body orientation tracking

The AI system detects human body posture and analyses body alignment using pose estimation techniques.

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2. Objective

The objective of this project was to:

- detect human body landmarks
- analyse body posture
- calculate body orientation angles
- detect body position
- improve AI skeletal tracking systems
- study pose estimation workflows

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3. Technologies Used

Technology	Purpose
Python	Programming
OpenCV	Image Processing
MediaPipe	Human Pose Detection

Artificial Intelligence	Pose Analysis
Computer Vision	Body Tracking

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4. Features Implemented

The AI system successfully implemented:

- human landmark detection
- skeletal tracking
- body angle detection
- body position estimation
- posture analysis
- AI pose visualization

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5. Workflow of the System

Step 1 — Human Image Input

The AI system accepts uploaded human body images for posture analysis.

Step 2 — Human Landmark Detection

MediaPipe Pose Estimation detects:

- shoulders
- elbows
- wrists
- hips
- knees
- ankles

using AI body tracking.

Step 3 — Coordinate Extraction

The AI system extracts:

- x coordinates
- y coordinates

from body landmarks.

Step 4 — Angle Calculation

The body angle is calculated using landmark coordinate analysis.

The angle calculation helps analyse:

- body tilt
- shoulder alignment
- body orientation
- posture direction

The body angle is calculated using:

$$\theta = \tan^{-1}\left(\frac{y_2 - y_1}{x_2 - x_1}\right)$$

-10-8-6-4-2246810-10-5510A(-8, -8)B(8, 8)m = 1.00

Step 5 — Position Detection

The AI system detects:

- standing straight posture
- tilted posture
- body orientation
- posture alignment

using skeletal tracking.

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6. Code Implementation:

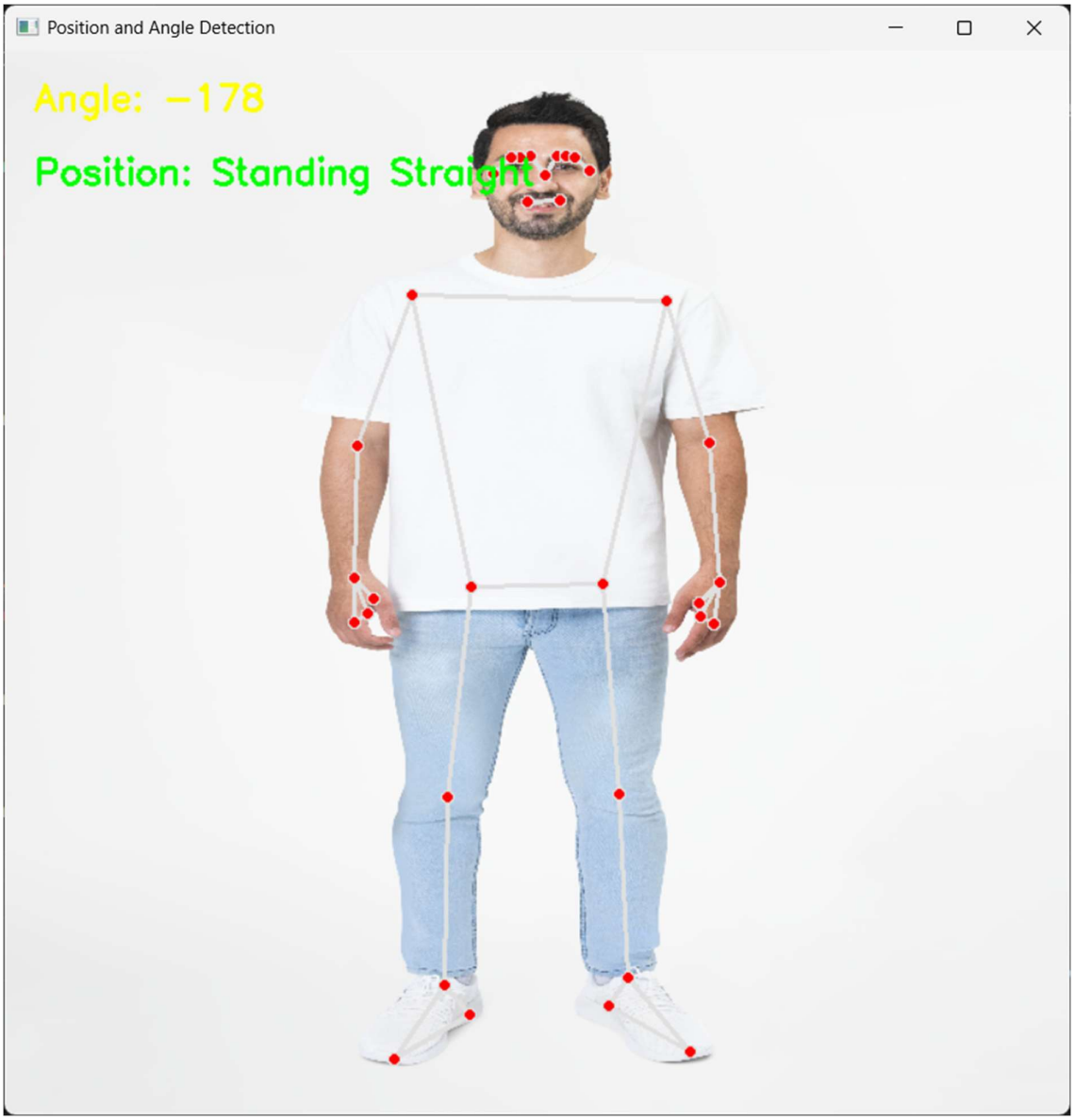
```python
import cv2
import mediapipe as mp
import math

image = cv2.imread("human.jpg")

if image is None:
    print("Image not found")
    exit()

image = cv2.resize(image, (720, 720))

mp_pose = mp.solutions.pose
mp_draw = mp.solutions.drawing_utils

pose = mp_pose.Pose(
    static_image_mode=True,
    min_detection_confidence=0.7
)

rgb = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)

results = pose.process(rgb)

if results.pose_landmarks:
```

```python

    lm = results.pose_landmarks.landmark

    h, w, _ = image.shape

    ls = lm[11]
    rs = lm[12]

    x1, y1 = int(ls.x*w), int(ls.y*h)
    x2, y2 = int(rs.x*w), int(rs.y*h)

    angle = int(
        math.degrees(
            math.atan2(y2-y1, x2-x1)
        )
    )

    if abs(y1-y2) < 20:
        position = "Standing Straight"

    elif y1 > y2:
        position = "Tilted Right"

    else:
        position = "Tilted Left"
```

```
51
52      mp_draw.draw_landmarks(
53          image,
54          results.pose_landmarks,
55          mp_pose.POSE_CONNECTIONS
56      )
57
58      cv2.putText(
59          image,
60          f"Angle: {angle}",
61          (20,40),
62          cv2.FONT_HERSHEY_SIMPLEX,
63          0.8,
64          (0,255,255),
65          2
66      )
67
68      cv2.putText(
69          image,
70          f"Position: {position}",
71          (20,90),
72          cv2.FONT_HERSHEY_SIMPLEX,
73          0.8,
74          (0,255,0),
75          2
76      )
```

```
75          2
76      )
77
78      print("Angle:", angle)
79      print("Position:", position)
80
81  else:
82      print("No Human Detected")
83
84  cv2.imwrite("output.jpg", image)
85
86  cv2.imshow("Position and Angle Detection", image)
87
88  cv2.waitKey(0)
89
90  cv2.destroyAllWindows()
```

Output:



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7. Algorithm

Step 1

Load uploaded image using OpenCV.

Step 2

Detect human body landmarks using MediaPipe Pose.

Step 3

Extract body landmark coordinates.

Step 4

Calculate body orientation angle using shoulder coordinates.

Step 5

Analyse body posture and body position.

Step 6

Generate skeletal visualization and posture analysis.

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8. Pseudo Code

START

Load uploaded human image

Detect body landmarks

Extract body coordinates

Calculate body angle

Detect body position

Generate skeletal visualization

Display final posture analysis

END

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9. Applications

The AI system can be used in:

- fitness monitoring
- posture correction systems
- sports analysis
- rehabilitation systems
- AI motion tracking
- healthcare applications

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10. Advantages

The advantages include:

- accurate posture analysis
- AI body angle estimation
- skeletal visualization
- intelligent body tracking
- real-time pose estimation

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11. Challenges Faced

Challenge	Impact
Poor lighting	Reduces landmark accuracy
Side body posture	Affects angle detection
Fast movement	Reduces tracking stability
Complex background	Reduces pose accuracy

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12. Expected Output

The AI system generates:

- body landmark visualization
- skeletal tracking
- body angle estimation
- body posture analysis
- body position detection

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13. Learning Outcomes

During this project, the following concepts were understood:

- AI pose estimation
- skeletal visualization
- body posture analysis
- body angle calculation
- computer vision workflows

14. Conclusion

The AI-Based Body Position and Angle Detection System successfully demonstrated:

- AI human pose estimation
- body posture analysis
- skeletal tracking
- body angle detection
- intelligent body orientation analysis

using Artificial Intelligence and Computer Vision technologies.

The project improved understanding of:

- body landmark detection
- posture estimation systems
- skeletal visualization
- AI body tracking
- computer vision applications